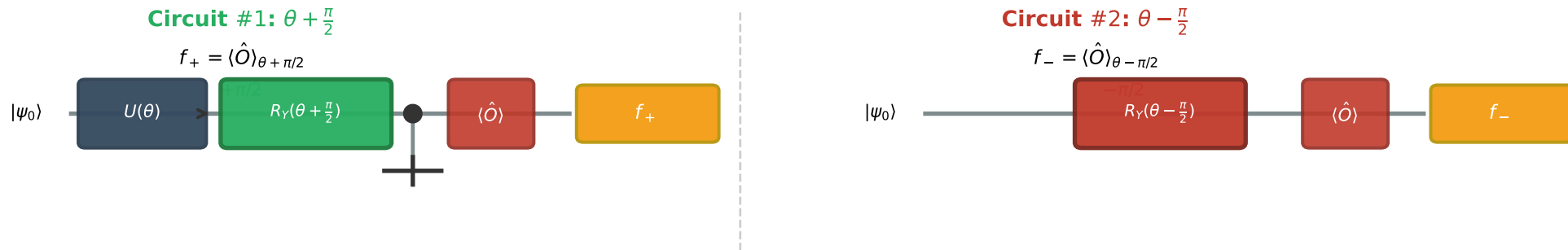


(a) Two Circuit Evaluations for Exact Gradient Computation



(b) Gradient Formula

$$\frac{\partial \langle \hat{O} \rangle}{\partial \theta_j} = \frac{1}{2} [f_+ - f_-]$$

*Exact analytic gradient
No finite differences needed!*

(c) Finite Difference (biased)

$$\frac{\partial \langle \hat{O} \rangle}{\partial \theta_j} \approx \frac{f(\theta + \epsilon) - f(\theta)}{\epsilon}$$

*Requires small ϵ
Introduces numerical bias*

(d) Key Insight

For $R_Y(\theta)$ gates:
 $R_Y(\theta) = e^{-i\theta\sigma_y/2}$
 σ_y has eigenvalues ± 1
 $\Rightarrow$ **Shift by $\pi/2$ swaps eigenstates**

Fig QC-3: Parameter-Shift Rule gradient computation. The exact analytic gradient $\partial \langle \hat{O} \rangle / \partial \theta$ is obtained from two circuit evaluations with shifted angles $\theta \pm \pi/2$, requiring no finite-difference approximation. The $\frac{1}{2}$ prefactor arises from the σ_y generator eigenvalues.